Organocatalysis

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The First Organocatalytic Cope Rearrangement

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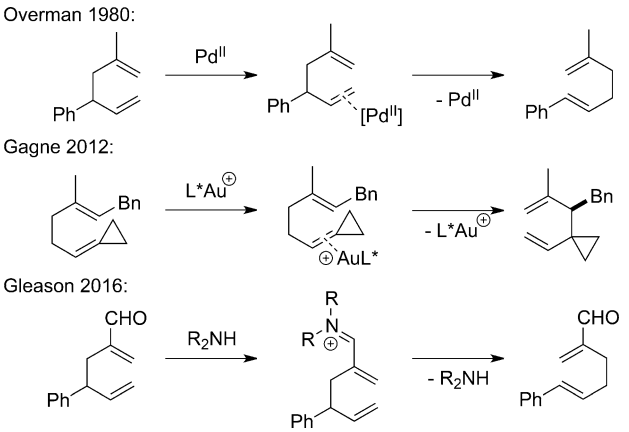
Cope rearrangements · diazepanes · iminium ions · organocatalysis · sigmatropic rearrangements

The classical Cope rearrangement of 1,5-hexadienes stands as a prominent example among [3,3]-sigmatropic rearrangements. It typically involves a stereospecific and concerted migration of an allyl group along a second allyl fragment within a 1,5-diene with concomitant σ- and π-bond reorganization.[1] As this transformation is principally a reversible reaction and often requires harsh reaction conditions, it has not become quite as popular as the related Claisen rearrangement in synthetic organic chemistry.[2]

A major landmark in the development of catalytic variants of the Cope rearrangement was the seminal work of Overman and co-workers who showed that this reaction can be catalyzed by Pd^{II} salts that are capable of π bonding to the diene (Scheme 1).[3] Since then, there has been only



Scheme 1. Metal-catalyzed (asymmetric) Cope rearrangements and the new organocatalytic activation mode.

a single example of a metal-catalyzed asymmetric Cope rearrangement, which was reported by the Gagne group in 2012 and based on the use of a chiral gold complex (Scheme 1).[4] Both the release of ring strain and the carbophilic nature of the gold catalyst were crucial for the success of this reaction.

A conceptually different approach for the activation of 1,5-dienes would rest upon the interaction of a catalyst not with the 1,5-diene core itself, but with substituents attached to it. It has been well established that carbonyl groups in the 2-position of a 1,5-diene exhibit a rate-accelerating effect on Cope rearrangements, which can be further enhanced with Lewis or Brønsted acids, resulting in the lowering of the LUMO energy.[5]

Based on this knowledge, Gleason and Kaldre envisioned that the reversible formation of an iminium ion from 1,5-hexadiene-2-carboxaldehydes might have the same effect and substantially lower the activation energy of the sigmatropic rearrangement (Scheme 1).[6] Furthermore, this approach would also provide the opportunity to design an enantioselective process with only catalytic quantities of a chiral amine. Conceptually, this idea was derived from the seminal work of MacMillan and co-workers who first established the capability of N-heterocycles to act as chiral catalysts for a range of enantioselective carbon–carbon bond-forming reactions of aldehydes and ketones.[7]

To verify this concept, the Cope rearrangement of 4-phenyl-1,5-hexadiene-2-carboxaldehyde (**1**), which furnishes the conjugated 1,5-hexadiene **2**, was selected as a model reaction owing to its favorable thermodynamic equilibrium (thermal rearrangement: toluene, 110 °C, 24 h, 90 %). Surprisingly, it turned out that primary amines were not capable of catalyzing the desired rearrangement. When secondary amines such as proline were employed as the catalyst, the Cope product was only formed in poor yields even at elevated temperatures. These findings are likely a result of slow iminium ion formation with the sterically hindered α-substituted enal. This lack of reactivity could be overcome by using more nucleophilic *N*-acyl hydrazides as catalysts (α-effect), which should facilitate iminium ion formation.[7] In fact, the conversion improved significantly with a selection of acyclic and cyclic acyl hydrazides. Careful catalyst and reaction optimization eventually led to an optimized procedure, which required only 10 mol % of the seven-membered *N*-acyl hydrazide **3** in combination with 10 mol % of triflic acid as co-catalyst in CH_3CN as the solvent. Under these conditions, the reaction proceeded to completion within 24 h at room temperature, and product **2** was isolated in 90 % yield (Scheme 2).

This method was subsequently applied to a range of 1,5-hexadienes, which were converted into the corresponding products with yields typically in the range of 70–90 %.

Not surprisingly, the N-methylated version of diazepane **3** did not catalyze the Cope rearrangement at all, corroborating the assumption of an iminium ion activation mode. The reason for the better performance of diazepane catalyst **3** compared to catalysts with smaller ring sizes has not been

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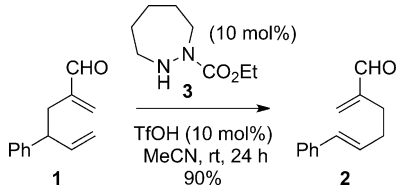
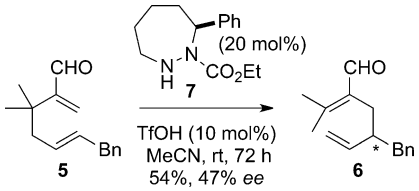


Scheme 2. Organocatalytic Cope rearrangement catalyzed by *N*-acyl diazepane **3**.

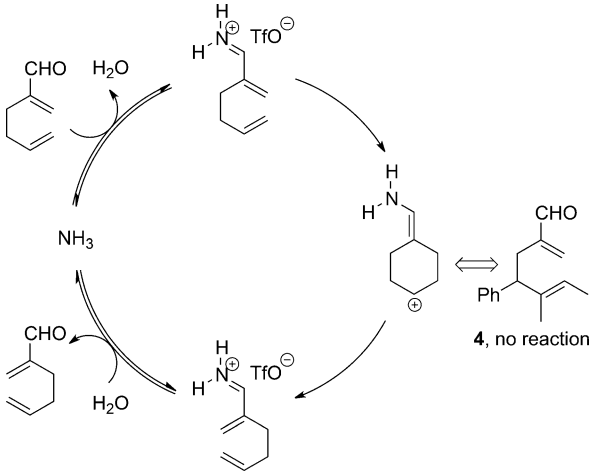


Scheme 4. Asymmetric variant of the organocatalytic Cope rearrangement.

definitively clarified at this time, but the authors speculated on a more rapid iminium ion formation based on calculated proton affinities. Moreover, independent DFT calculations for cinnamaldehyde as the substrate have shown that LUMO lowering is more effective for catalysts with larger rings.[8]

In order to elucidate the mechanism of this Cope rearrangement in more detail, the authors conducted preliminary DFT calculations (B3LYP/6-31*) on a simplified substrate and with ammonia as the catalyst. These calculations suggest a stepwise pathway proceeding through a low-energy carbocation intermediate rather than a concerted reaction mechanism. On that basis, the present reaction can be viewed as the conjugate addition of an alkene to an α,β-unsaturated iminium ion followed by fragmentation of the cyclohexyl carbocation (Scheme 3). This scenario somewhat resembles the cyclization-induced rearrangement mechanism that was proposed for the palladium(II)-catalyzed Cope rearrangement developed by Overman.[3]

In light of the theoretical findings, it is surprising, however, that the 5,6-disubstituted 1,5-hexadiene **4** failed to give the desired rearranged product under the established conditions. The positive charge developing at the C5 position of the predicted carbocation intermediate should be stabilized even more efficiently by a neighboring alkyl group. For the same reason, alkyl substitution at either the C2 or C5 position is a prerequisite for palladium(II)-catalyzed Cope rearrangements. The authors clearly recognized this aspect and pointed out that there is currently only computational evidence for the proposed stepwise nature of the reaction. Further experimental and theoretical studies are needed to unambiguously and firmly establish the mechanism of this new organocatalytic Cope rearrangement.



Scheme 3. Proposed catalytic cycle on the basis of DFT calculations.

In a further extension of this concept, the authors synthesized four new chiral diazepanes to evaluate their potential as catalysts for this reaction. With 20 mol % of chiral *N*-acyl diazepane **7**, the Cope rearrangement of 1,5-hexadiene **5** delivered product **6** in 54 % yield and 47 % *ee* (Scheme 4). While this result is only preliminary, it confirms the feasibility of this approach to be rendered enantioselective. It is almost certain that more active as well as enantioselective chiral *N*-acyl hydrazides will emerge in the future for this important class of reactions.

In conclusion, Gleason and Kaldre have developed the first and thus far only organocatalytic Cope rearrangement of all-carbon 1,5-hexadienes. By introducing a carbonyl group in the 2-position of the 1,5-diene and using *N*-acyl diazepane **3** as the catalyst, they were able to apply the concept of LUMO-lowering iminium ion catalysis to Cope rearrangements. This process was only successful with *N*-acyl hydrazides as catalysts, highlighting their unique potential to activate α-branched enals. These results should pave the way for further development in the field of organocatalytic sigmatropic rearrangements and should also stimulate more efforts towards the development of new chiral cyclic *N*-acyl hydrazides as a promising class of organocatalysts.

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Highlights

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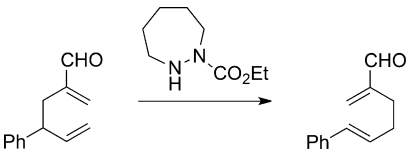

iminium ion catalysis

Exploiting the concept of iminium ion catalysis led to the discovery of the first organocatalytic Cope rearrangement of 1,5-hexadiene-2-carboxaldehydes. A diazepane catalyst exhibited optimal catalytic efficiency and offers possibilities for chiral modification.

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